Richard Blazek

moi@richardblazek.com | +353 87 439 3163 | richardblazek.com | GitHub | LinkedIn

Education

Integrated Master in Computer Science (MCS) — Trinity College Dublin Sep 2022 – May 2027 (expected)

- Overall grade: I (First class Honours)
- Full-year exchange in Hong Kong (HKUST) in 2024/25
- Thesis: *Looking at Each Other: A Dataset of Social Gaze in Conversation* (supervisor Prof. Rachel McDonnell)

Work Experience

NPU Compiler Intern — Intel, Ireland Jan 2026 – Jul 2026

- Built an analysis tool for the MLIR-based (C++) compiler that detected **62** obsolete compiler passes
- Implemented passes for a host pipeline that speeds up compiling models with repeating blocks (e.g. LLaMA)
- Made the test suite **3× faster** by lazily constructing an expensive object

Machine Learning Research Intern — Heidelberg University, Germany May 2024 – Aug 2024

- Reduced training time of a CNN pipeline for mammography image segmentation by **83%**
- Implemented SMOTE and built a pipeline handling an imbalanced dataset of six tumour types
- Added hyperparameters for adjusting the split between undersampling, SMOTE and class weighting

Full-Stack Developer — MicroDoc.io, Ireland Sep 2022 – Aug 2023

- Redesigned the onboarding flow, built the medical transcription page and Gmail/Outlook integration
- Containerised the app with Docker and migrated it from Google Cloud to AWS
- Secured the Python backend by implementing endpoint-level access control and data encryption

Projects

Programming language compiler

- Designed a statically-typed structured programming language
- Built a compiler in **Rust**, compiling my language into **WebAssembly**
- Prototype available online: richardblazek.com/zyba-wasm-compiler.html

Activities

Democracy Hackathon — Council of Europe, France 2026

- Selected as one of **20 teams** out of **120 applicant teams** from 46 member states
- Built a system for detecting coordinated inauthentic behaviour (CIB) in social networks
- Used narrative clustering to discover groups from the data and timing patterns to detect coordination

Patch (startup accelerator) — Dogpatch Labs, Ireland 2024

- Led a team of three people, building an employment platform for people with disabilities
- Interviewed HR managers at Stripe and representatives from disability NGOs

Skills

Languages: C++, Python, Rust, Go, SQL, JavaScript
Compilers: MLIR, LLVM, WebAssembly
Machine Learning: PyTorch, NumPy, Pandas, SciPy
Infrastructure: Docker, Linux, Bash, Git, PostgreSQL, MySQL, Amazon Web Services (AWS)